Strategic Reasoning: From beliefs about mistakes to mistaken beliefs

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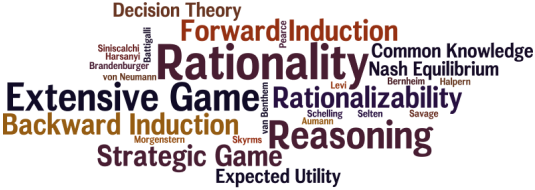

Plan

1. Motivation and introduction: Epistemic game theory
2. A model of rational deliberation in strategic games
3. Deliberating between backward and forward induction
4. Concluding remarks

The Epistemic Foundations of Game Theory

...the analysis constitutes a fleshing-out of the textbook interpretation of equilibrium as ‘rationality plus correct beliefs.’

E. Dekel and M. Siniscalchi. *Epistemic Game Theory*. Handbook of Game Theory with Economic Applications, 2015.

A. Brandenburger. *The Language of Game Theory: Putting Epistemics into the Mathematics of Games*. World Scientific, 2014.

A. Perea. *Epistemic Game Theory: Reasoning and Choice*. Cambridge University Press, 2012.

EP and O. Roy. *Epistemic Game Theory*. Stanford Encyclopedia of Philosophy, 2015.

Rational Choice

$$EU(A) = \sum_{o \in O} P_A(o) \times U(o)$$

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how likely it is that o will occur, supposing that A is chosen.

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R. A. Briggs. *Normative Theories of Rational Choice: Expected Utility*. Stanford Encyclopedia of Philosophy, 2014, plato.stanford.edu/entries/rationality-normative-utility/.

Rational Choice: Decision Problem

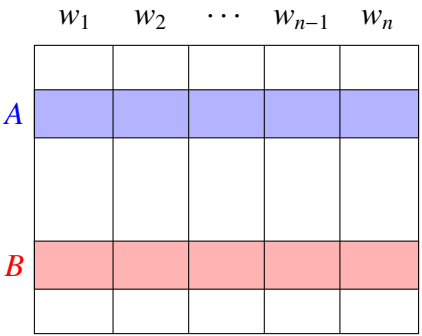
A

B

Rational Choice: Decision Problem

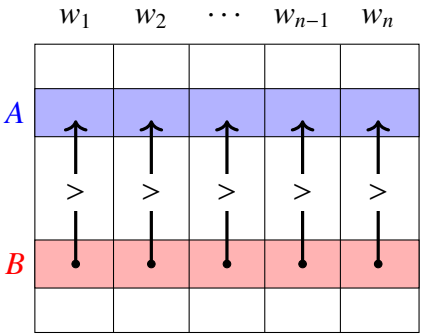
	w_1	w_2	$\cdots$	w_{n-1}	w_n
A					
B					

Rational Choice: Decision Problem



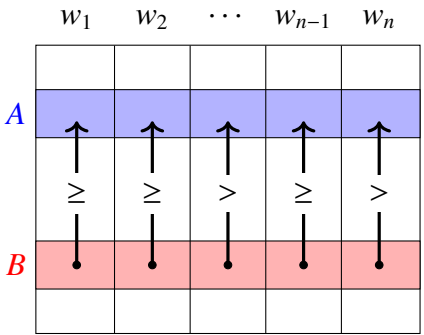
An **act** is a function $F : W \to O$

Rational Choice: Strict Dominance



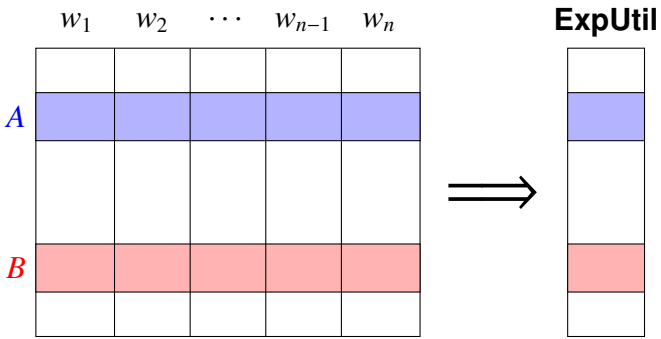
$$\forall\ w \in W,\ u(A(w)) > u(B(w))$$

Rational Choice: Weak Dominance



$\forall\ w \in W,\ u(A(w)) \geq u(B(w))$ and $\exists\ w \in W,\ u(A(w)) > u(B(w))$

Rational Choice: Maximize (Subjective) Expected Utility



$$\sum_{w \in W} P_A(w) * u(A(w)) > \sum_{w \in W} P_A(w) * u(B(w))$$

Just Enough Game Theory

A **game** is a mathematical model of a strategic interaction that includes

- a set of decision makers (called players)
- the available actions for each player
- the players' interests (i.e., preferences),
- the "structure" of the decision problem

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- a set of decision makers (called players)
- the available actions for each player
- the players' interests (i.e., preferences),
- the "structure" of the decision problem

It does not specify the actions that the players do take.

Games

		B	
		l	r
A	u	3, 1	0, 0
	d	0, 0	1, 3

Games

		B	
		l	r
A	u	3, 1	0, 0
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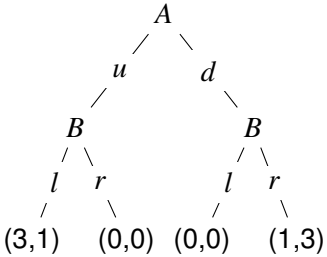

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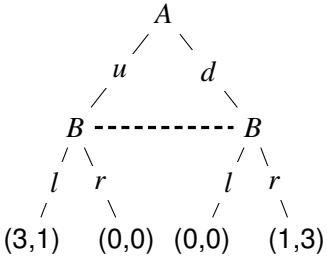

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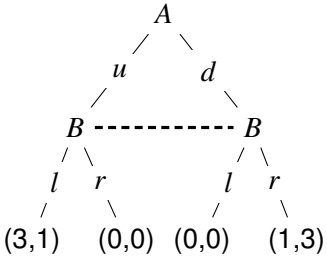
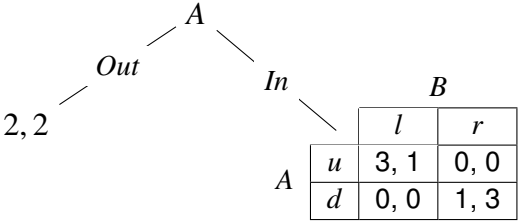




From Decisions to Games

"*[T]he* fundamental insight of game theory [is] that a rational player must take into account that the players reason about each other in deciding how to play."

R. Aumann and J. Dreze. *Rational Expectations in Games*. American Economic Review, 98, pp. 72-86, 2008.

Solution Concept

A **solution concept** is a systematic description of the outcomes that may emerge in a family of games.

This is the starting point for most of game theory and includes many variants: Nash equilibrium, backwards induction, or iterated dominance of various kinds.

These are usually thought of as the embodiment of “rational behavior” in some way and used to analyze game situations.

Let $G = \langle (S_i)_{i \in N}, (u_i)_{i \in N} \rangle$ be a finite strategic game (each S_i is finite and the set of players N is finite).

A **strategy profile** is an element $\sigma \in S = S_1 \times \cdots \times S_n$

σ is a (pure strategy) **Nash equilibrium** provided for all i, for all $s_i \in S_i$,

$$u_i(\sigma) \geq u_i(s_i, \sigma_{-i})$$

Matching Pennies

		Bob	
		H	T
Ann	H	1,-1	-1, 1
	T	-1,1	1,-1

There are no pure strategy Nash equilibria.

Mixed Strategies

		Bob	
		H	T
Ann	H	1,-1	-1, 1
	T	-1,1	1,-1

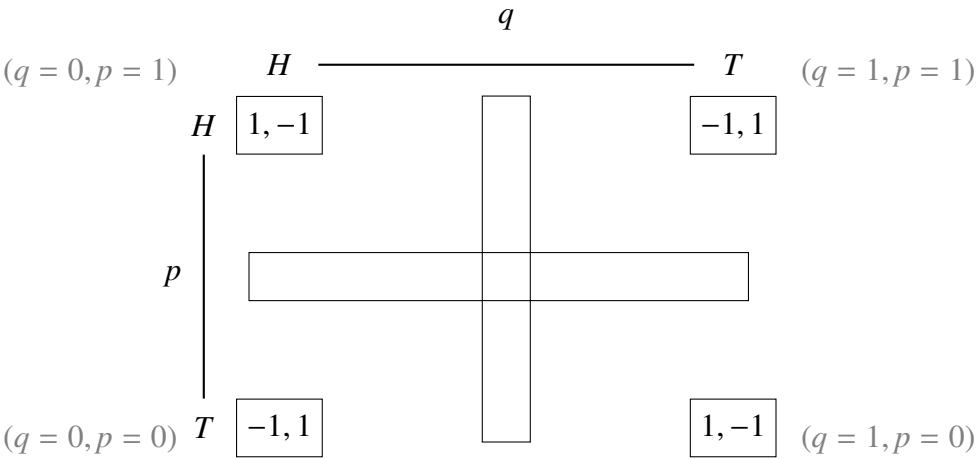
A **mixed strategy** is a probability distribution over the set of pure strategies. For instance:

- $[1/2 : H, 1/2 : T]$
- $[1/3 : H, 2/3 : T]$
- ...

Mixed Extension

$(q = 0, p = 1)$		H	T	$(q = 1, p = 1)$
	H	$\boxed{1, -1}$	$\boxed{-1, 1}$	
$(q = 0, p = 0)$	T	$\boxed{-1, 1}$	$\boxed{1, -1}$	$(q = 1, p = 0)$

Mixed Extension



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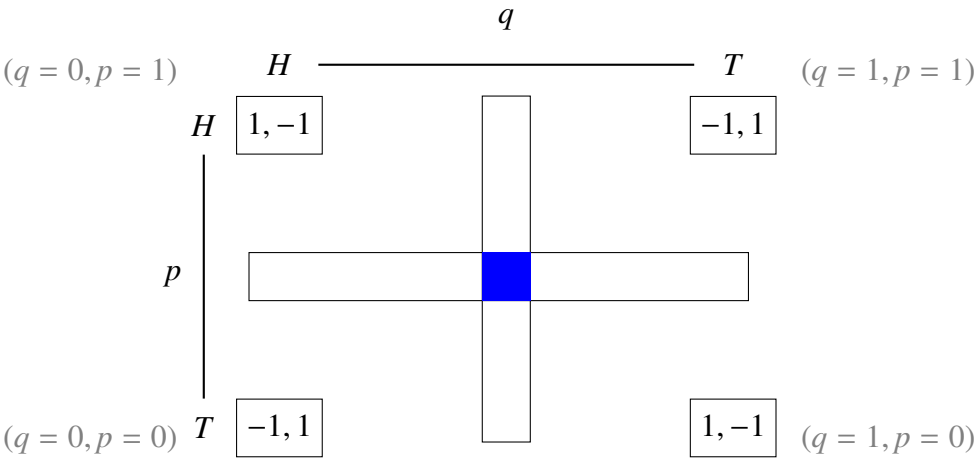

$$p(1-q) - pq - (1-p)(1-q) + (1-p)q, \; -p(1-q) + pq + (1-p)(1-q) - (1-p)q$$

“...no, equilibrium is not the way to look at games. Now, Nash equilibrium is king in game theory. Absolutely king. We say: No, Nash equilibrium is an interesting concept, and it’s an important concept, but it’s not the most basic concept. The most basic concept should be: to maximise your utility given your information. It’s in a game just like in any other situation. Maximise your utility given your information!”

Robert Aumann, 5 Questions on Epistemic Logic, 2010

From Nash to rationalizability...

Bob

Ann	L	C	R
T	3, 2	0, 0	2, 3
M	0, 0	1, 1	0, 0
B	2, 3	0, 0	3, 2

		Bob		
		L	C	R
Ann	T	3, 2	0, 0	2, 3
	M	0, 0	1, 1	0, 0
	B	2, 3	0, 0	3, 2

(M, C) is the unique Nash equilibrium

		Bob		
		L	C	R
Ann	T	3, 2	0, 0	2, 3
	M	0, 0	1, 1	0, 0
	B	2, 3	0, 0	3, 2

T, L, B and R are **rationalizable**

		Bob		
		L	C	R
Ann	T	3, 2	0, 0	2, 3
	M	0, 0	1, 1	0, 0
	B	2, 3	0, 0	3, 2

T, L, B and R are **rationalizable**

		Bob		
		L	C	R
Ann	T	3, 2	0, 0	2, 3
	M	0, 0	1, 1	0, 0
	B	2, 3	0, 0	3, 2

Ann plays B because she thought Bob will play R

		Bob		
		L	C	R
Ann	T	3, 2	0, 0	2, 3
	M	0, 0	1, 1	0, 0
	B	2, 3	0, 0	3, 2

Bob plays L because he thought Ann will play B

Bob

Ann	L	C	R
T	3, 2	0, 0	2, 3
M	0, 0	1, 1	0, 0
B	2, 3	0, 0	3, 2

Bob was correct, but Ann was wrong

Bob

Ann	L	C	R	X
T	3, 2	0, 0	2, 3	0, -5
M	0, 0	1, 1	0, 0	200,-5
B	2, 3	0, 0	3, 2	1,-3

Not every strategy is rationalizable: Ann can't play M *because* she thinks Bob will play X

“Analysis of strategic economic situations requires us, implicitly or explicitly, to maintain as plausible certain psychological hypotheses. The hypothesis that real economic agents universally recognize the salience of Nash equilibria may well be less accurate than, for example, the hypothesis that agents attempt to “out-smart” or “second-guess” each other, believing that their opponents do likewise.” (pg. 1010)

B. D. Bernheim. *Rationalizable Strategic Behavior*. Econometrica, 52:4, pgs. 1007 - 1028, 1984.

		B	
		L	R
A	U	3,1	0,0
	D	0,0	1,3

		B	
		L	R
A	U	3,1	0,0
	D	0,0	1,3

		$Play_B(L)$	$Play_B(R)$
A	U	3	0
	D	0	1

		$Play_A(U)$	$Play_A(D)$
B	L	1	0
	R	0	3

Beliefs, or conjectures

Suppose that $G = \langle (S_i)_{i \in N}, (u_i)_{i \in N} \rangle$ is a game.

- Outcomes of the game: $S = \Pi_{i \in N} S_i$
- State of uncertainty for i: $S_{-i} = \Pi_{j \in N, j \neq i} S_j$
- Player i's **beliefs** (or **conjectures**): $P_i \in \Delta(S_{-i})$

  $\Delta(X)$ *is the set of probabilities measures over* X

Epistemic Game Theory

- Compare and contrast the different representations of knowledge and beliefs.

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- What are the implications of assuming *common* belief (knowledge, strong belief, assumption, etc.) of rationality?

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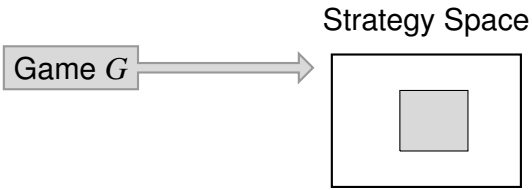
  What are the epistemic assumptions underlying well-known solution concepts (e.g, Nash equilibrium, correlated equilibrium, iterated strict dominance, iterated weak dominance)?
- What should players think about the joint actions of their opponents? (E.g., are the opponents' actions independent or correlated?)

The Epistemic Program in Game Theory

Game G

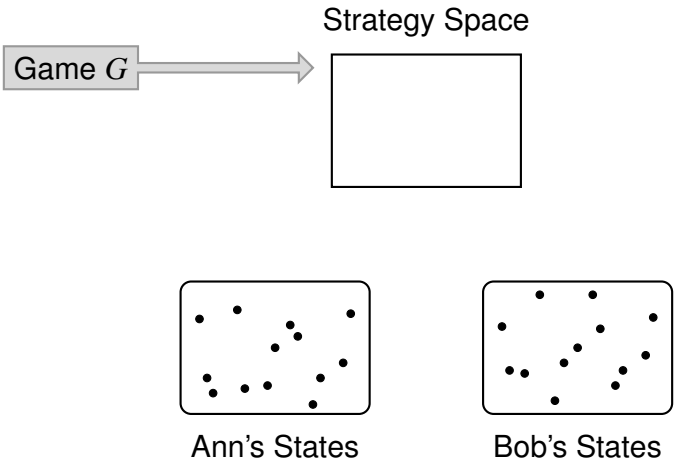
G: available actions, payoffs, structure of the decision problem

The Epistemic Program in Game Theory



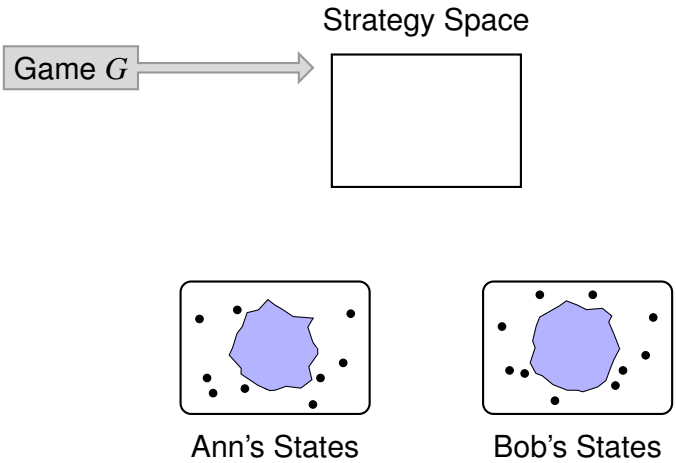
solution concepts are systematic descriptions of what players *do*

The Epistemic Program in Game Theory



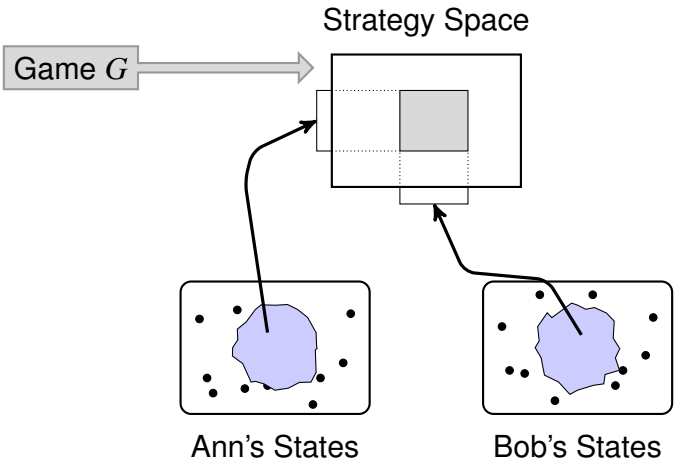
The game model includes *information states* of the players

The Epistemic Program in Game Theory



Restrict to information states satisfying some rationality condition

The Epistemic Program in Game Theory



Project onto the strategy space

“Intuitively, an epistemic model of a game can be interpreted as representing the players’ reasoning. Indeed, before making a decision in a game, a player reasons about the game and his opponents, given his knowledge and beliefs. Precisely these epistemic mental states on which a player bases his decisions and which characterize his reasoning are described in an epistemically enriched game-theoretic framework.”

C. Bach and A. Perea. *Utility proportional beliefs*. International Journal of Game Theory, 2014, 43, pgs. 881 - 902.

Epistemic Game Theory: Strategic Reasoning

- What are the assumptions about *strategic reasoning* that drive the epistemic analyses?

- What do the epistemic analyses tell us about the nature of rational choice in strategic situations?

Epistemic Game Theory: Strategic Reasoning

"The word *eductive* will be used to describe a dynamic process by means of which equilibrium is achieved through careful reasoning on the part of the players. Such reasoning will usually require an attempt to simulate the reasoning processes of the other players. Some measure of pre-play communication is therefore implied, although this need not be explicit. To reason along the lines "if I think that he thinks that I think..." requires that information be available on how an opponent thinks." (pg. 184)

K. Binmore. *Modeling Rational Players*. Economics and Philosophy, 3,179 - 21, 1987.

Deliberation in Games

- The Harsanyi-Selten tracing procedure
- Brian Skyrms' model of "dynamic deliberation"
- Ken Binmore's analysis using Turing machines to "calculate" the rational choice
- Robin Cubitt and Robert Sugden's "reasoning based expected utility procedure"
- Johan van Benthem et col.'s "virtual rationality *announcements*"

Different frameworks, common thought: *the "rational solutions" of a game are the result of individual deliberation about the "rational" action to choose.*

Rational deliberation in games

B. Skyrms (1990). *The Dynamics of Rational Deliberation*. Harvard University Press.

> *It is not just a question of what common knowledge obtains at the moment of truth, but also how common knowledge is preserved, created, or destroyed in the deliberational process which leads up to the moment of truth.* *(pg. 159)*

Deliberational Decision Theory

F. Arntzenius. *No Regrets, or: Edith Piaf Revamps Decision Theory*. Erkenntnis, 68, pgs. 277 - 297, 2008.

J. Joyce. *Regret and Instability in Causal Decision Theory*. Synthese, 187: 1, pgs. 123 - 145, 2012.

I. Douven. *Decision theory and the rationality of further deliberation*. Economics and Philosophy, 18, pgs. 303 - 328, 2002.

Deliberational Decision Theory

Current Evaluation: If P_t characterizes your beliefs at time t, then at t you should *evaluate* each act by its (causal, evidential) expected utility computed using P_t.

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Sometimes initial opinions fix actions, *but not always* (e.g., Murder Lesion, Psychopath Button)

Information Feedback

In the simplest case, deliberation is trivial; one calculates expected utility and maximizes

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Information feedback: “the very process of deliberation may generate information that is relevant to the evaluation of the expected utilities. Then, processing costs permitting, a Bayesian deliberator will feed back that information, modifying his probabilities of states of the world, and recalculate expected utilities in light of the new knowledge.” (Skyrms)

		B	
		L	R
A	U	3,1	0,0
	D	0,0	1,3

Strategic Game

		B	
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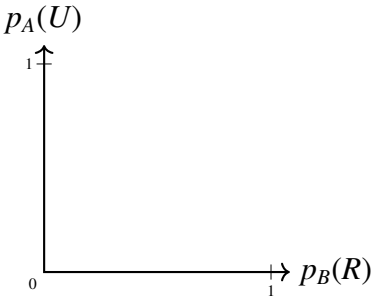
Strategic Game



Solution Space

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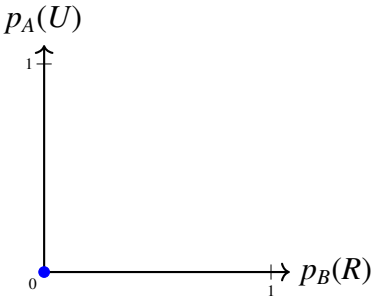
Strategic Game



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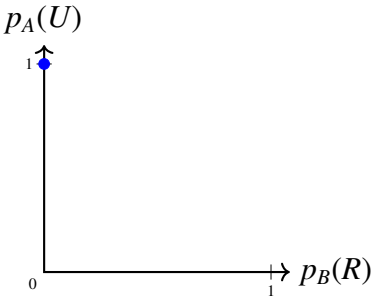

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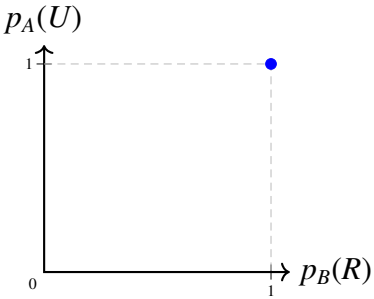

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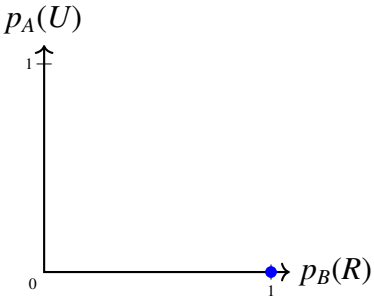

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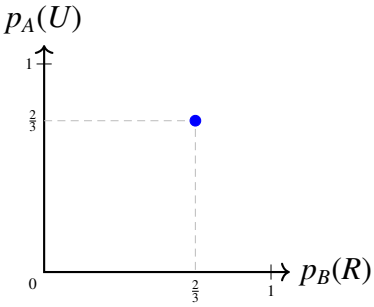

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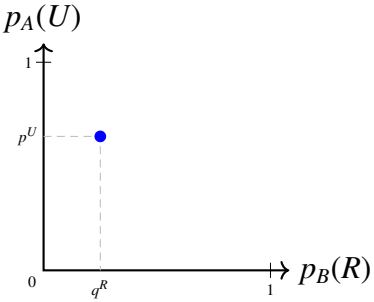
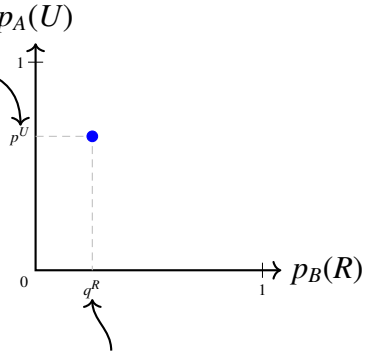


A's current state of indecision

		B	
		L	R
A	U	3,1	0,0
	D	0,0	1,3



A's belief about what B will do

B's belief about what A will do

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		L	R
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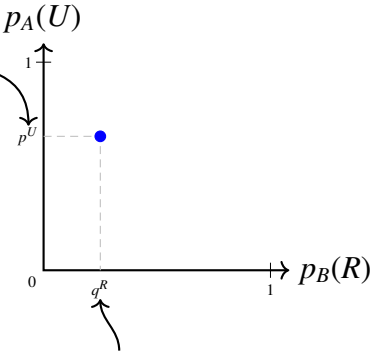
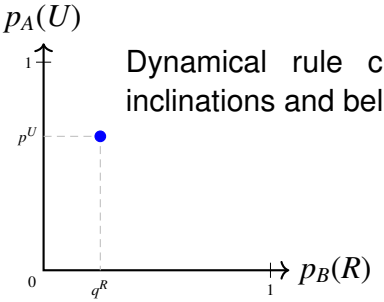
	D	0,0	1,3



B's current state of indecision

		B	
		L	R
A	U	3,1	0,0
	D	0,0	1,3



Dynamical rule changing inclinations and beliefs

$$G = \langle N, (S_i)_{i \in N}, (u_i)_{i \in N} \rangle$$

$$G = \langle N, (S_i)_{i \in N}, (u_i)_{i \in N} \rangle$$

For each player $i \in \mathcal{A}$, the **state of indecision** is a pair (I_i, P_i), where $I_i \in \Delta(S_i)$ is called i's **inclinations** and $P_i \in \Delta(S_{-i})$ is i's **beliefs** about the other player's choice.

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The **expected utility** of $s \in S_i$ is: $EU_i(s) = \sum_{t \in S_{-i}} P_i(t) u_i(s, t)$.

$$G = \langle N, (S_i)_{i \in N}, (u_i)_{i \in N} \rangle$$

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The **expected utility** of $s \in S_i$ is: $EU_i(s) = \sum_{t \in S_{-i}} P_i(t) u_i(s, t)$.

The **status quo** is: $SQ_i = \sum_{s_i \in S_i} I_i(s_i) EU_i(s_i)$.

Nash dynamics

The **covetability** of a strategy s for player i is: $cov_i(s) = \max(EU_i(s) - SQ_i, 0)$.

Nash dynamics

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Then, **Nash dynamics** rule transforms $I_i \in \Delta(S_i)$ into a new probability $I'_i \in \Delta(S_i)$ as follows. For each $s \in S_i$:

$$I'_i(s) = \frac{k \cdot I_i(s) + cov_i(s)}{k + \sum_{s \in S_i} cov_i(s)},$$

where $k > 0$ is the “index of caution”.

Bayes dynamics

The **Bayes dynamics**, also called **Darwin dynamics**, transforms $I_i \in \Delta(S_i)$ into a new probability $I_i' \in \Delta(S_i)$ as follows. For each $s \in S_i$:

$$I_i'(s) = I_i(s) + \frac{1}{k} I_i(s) \frac{EU_i(s) - SQ_i}{SQ_i}.$$

where $k > 0$ is the “index of caution”.

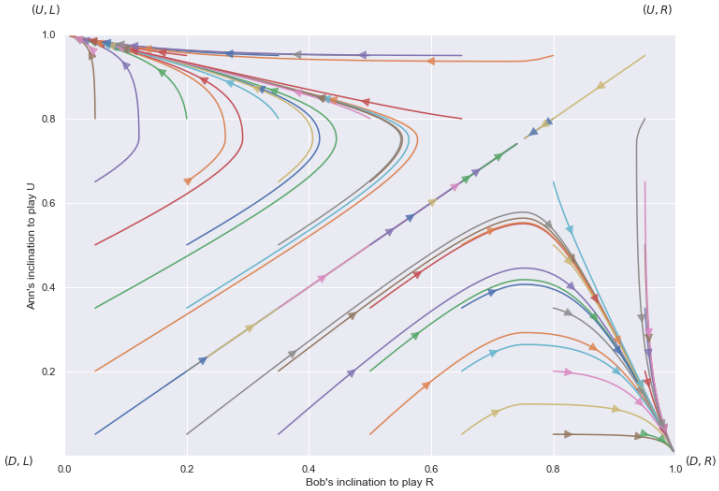
Update by emulation

1. The players' initial states of indecision and the dynamical rule used to update inclinations are common knowledge.
2. Each player assumes that the other players are rational deliberators who have just carried out a similar process. So, she can simply go through their calculations to see their new states of indecision and update her beliefs for their acts accordingly.

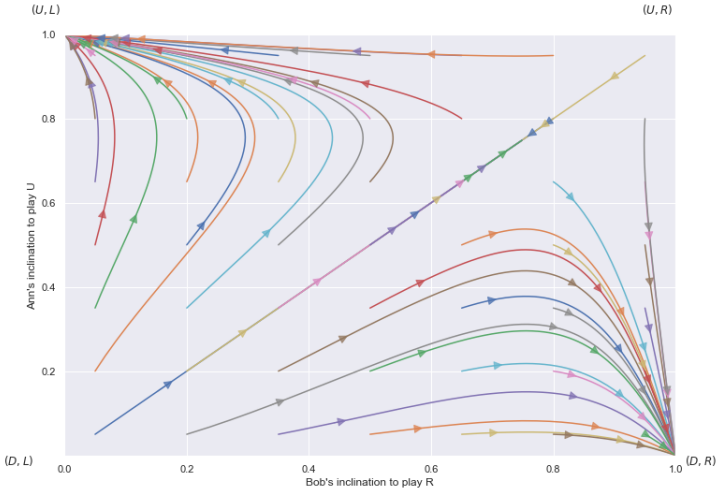
Battle of the sexes

		B	
		L	R
A	U	3,1	0,0
	D	0,0	1,3

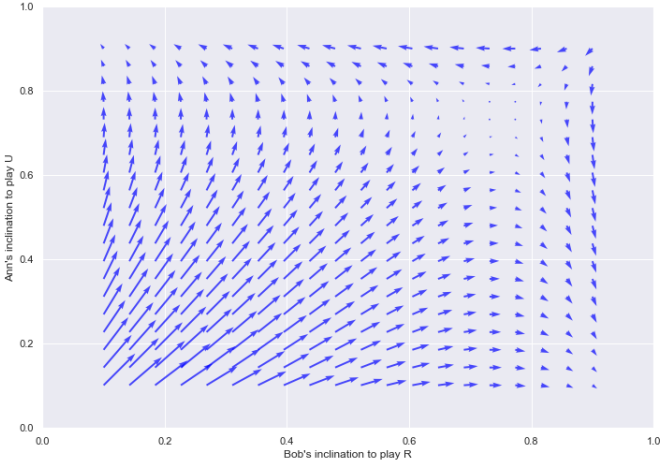
BoS: Nash deliberators



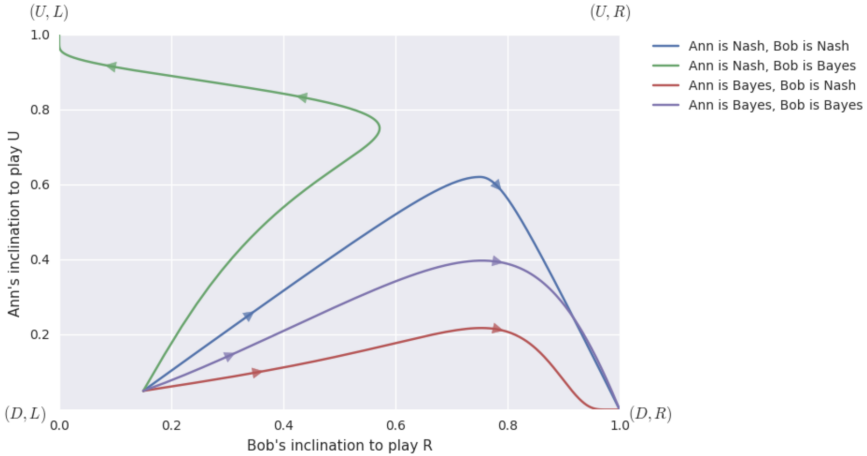
BoS: Bayes deliberators



BoS: Bayes deliberators - Phase Diagram



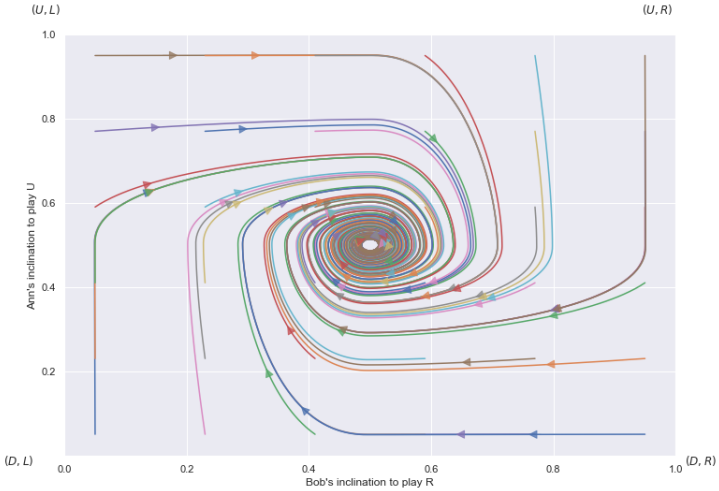
BoS: Comparing Deliberators



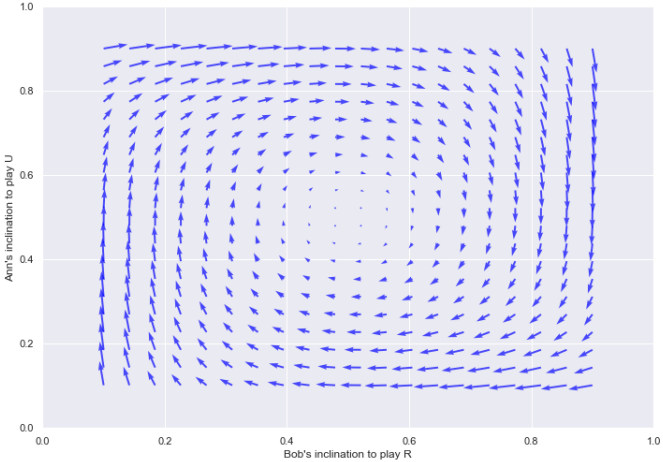
Matching pennies

		B	
		L	R
A	U	1,-1	-1,1
	D	-1,1	1,-1

Matching pennies - Nash deliberators



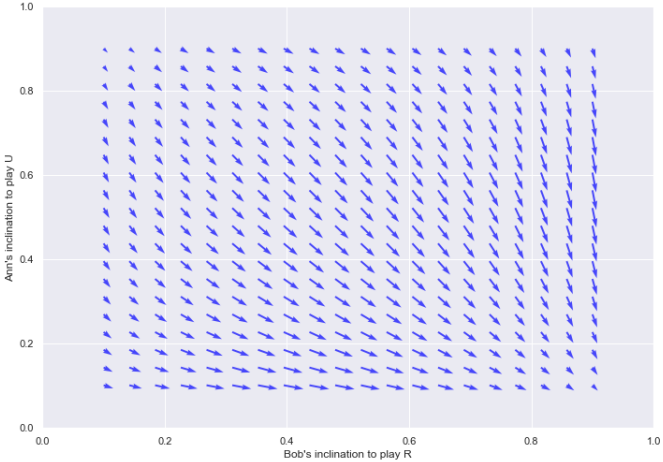
Matching pennies - Bayes deliberators



Prisoner's dilemma

		B	
		L	R
A	U	3,3	0,4
	D	4,0	1,1

Prisoner’s dilemma - Bayes deliberators



Moving away from update by emulation...

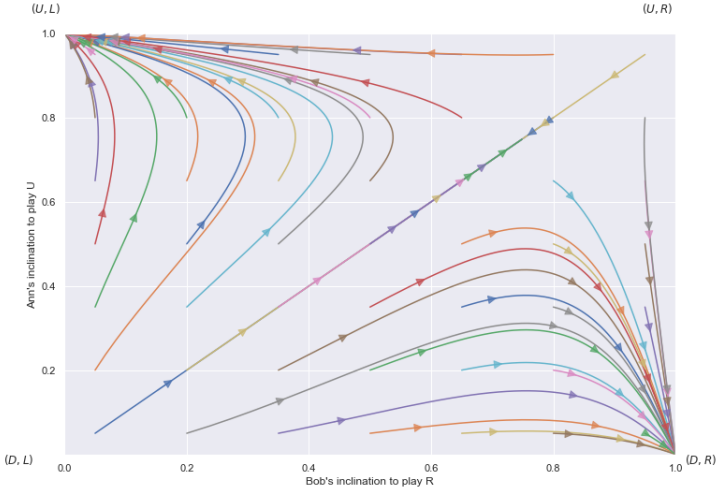
Moving away from update by emulation...

- Beliefs do not change during deliberation
- Pre-deliberational reasoning about what the opponent is going to do
- Radical uncertainty about the opponent's initial inclinations

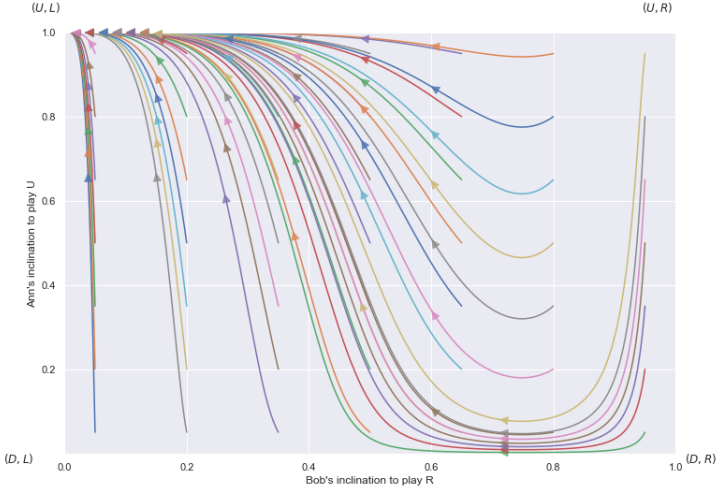
BoS

		B	
		L	R
A	U	3,1	0,0
	D	0,0	1,3

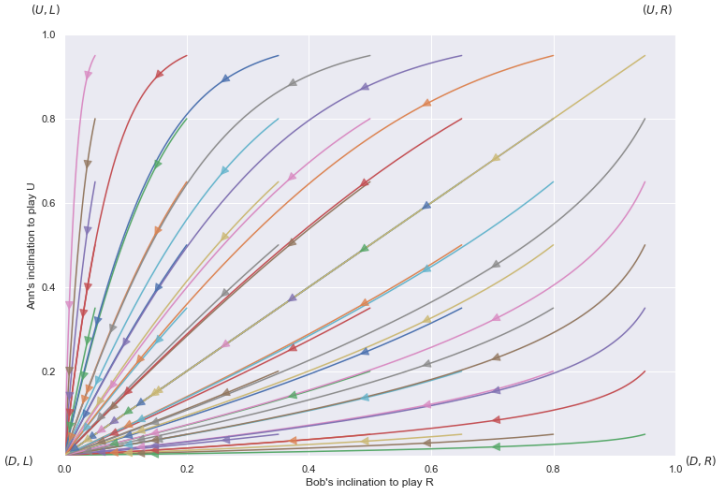
BoS - Update by emulation



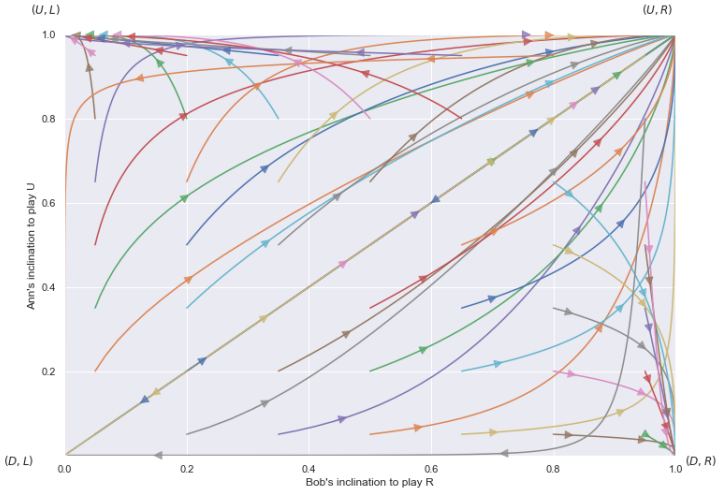
BoS: $P_B(U) = 0.8$


(U, L)
(U, R)
(D, L)
(D, R)
1.0
0.8
0.6
0.4
0.2
0.0
0.2
0.4
0.6
0.8
1.0
Ann's inclination to play U
Bob's inclination to play R

BoS: $P_B(U) = 0.9$, $P_A(R) = 0.9$



BoS: Fixed beliefs



Plan

- ✓ Motivation and introduction: Epistemic game theory
- ✓ A model of rational deliberation in strategic games
3. Deliberating between backward and forward induction
4. Concluding remarks

EP and A. Knoks. *Deliberating between Backward and Forward Induction Reasoning: First Steps*. proceedings of TARK, 2015.

EP and A. Knoks. *Deliberational dynamics in context*. Manuscript, 2018.

Motivation: Backward and forward induction reasoning

We think of backward and forward induction as styles of reasoning in extensive games:

Backward induction reasoning: player's ignore past behavior and reason only about their opponents' future moves.

Forward induction reasoning: player's rationalize past behavior and use it as a basis to form beliefs about their opponents' future moves.

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Forward induction reasoning: player's rationalize past behavior and use it as a basis to form beliefs about their opponents' future moves.

How do the players "choose" between the different styles of reasoning?

Motivation: Backward and forward induction reasoning

We do not mean to suggest that players explicitly *choose* how they reason in an extensive game.

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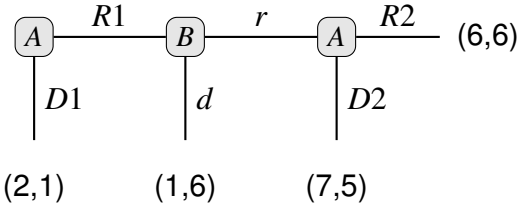
We do not mean to suggest that players explicitly *choose* how they reason in an extensive game.

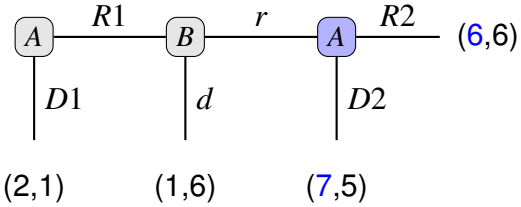
Our work is in line with the many epistemic characterization of backward and forward induction (e.g., Aumann (1995), Stalnaker (1998), Halpern (2001), Bonanno (2014), Perea (2010, 2014), Battigalli and Sinischalchi (2002), among others):

In epistemic models of extensive form games, the players optimal choices in the game are derived from their beliefs about what their opponents are going to do, higher-order beliefs (belief about what their opponents might be thinking), and belief revision policies.

BI Puzzle



BI Puzzle



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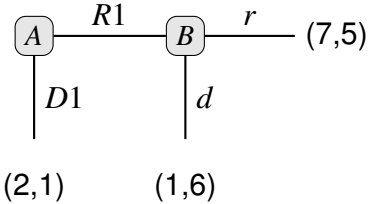

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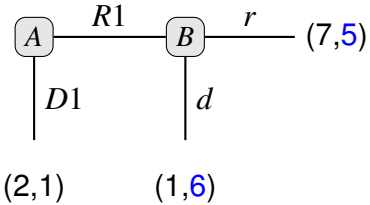

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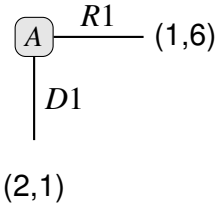

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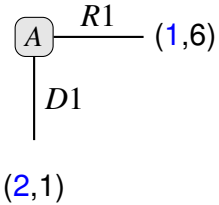

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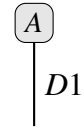
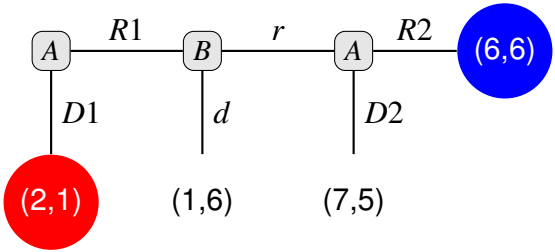
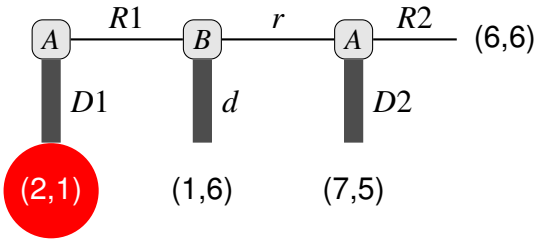


(2,1)

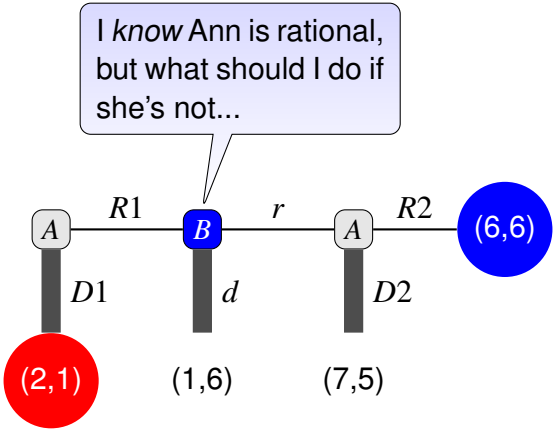
BI Puzzle



BI Puzzle?



BI Puzzle?


I *know* Ann is rational, but what should I do if she's not...
A
R1
B
r
A
R2
(6,6)
D1
d
D2
(2,1)
(1,6)
(7,5)

R. Aumann. *Backwards induction and common knowledge of rationality*. Games and Economic Behavior, 8, pgs. 6 - 19, 1995.

R. Stalnaker. *Knowledge, belief and counterfactual reasoning in games*. Economics and Philosophy, 12, pgs. 133 - 163, 1996.

J. Halpern. *Substantive Rationality and Backward Induction*. Games and Economic Behavior, 37, pp. 425-435, 1998.

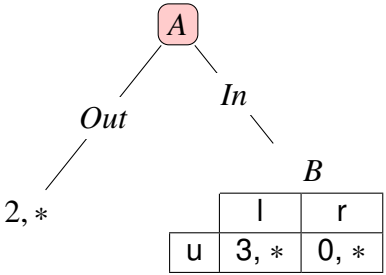
Informal characterizations of BI

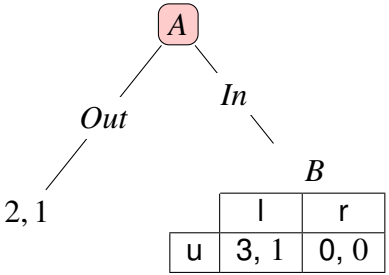
- Future choices are *epistemically independent* of any observed behavior
- Any “off-equilibrium” choice is interpreted simply as a mistake (which will not be repeated)
- At each choice point in a game, the players only reason about future paths

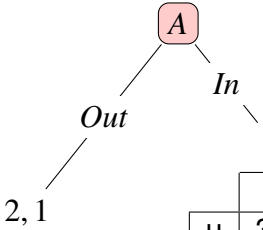
Rationalizing Observed Actions

After observing an (unexpected) move by some player, you could:

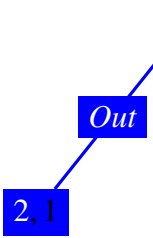
1. Change your belief about the player's rationality, but maintain your beliefs about the player's *passive beliefs*.
2. Change your belief about the player's passive beliefs, but maintain your belief in the player's rationality.
3. Conclude that the player perceives the game differently.


A
Out
In
2, ∗
B
l
r
u
3, ∗
0, ∗


A
Out
In
2, 1
B
l
r
u
3, 1
0, 0



		B	
		l	r
A	u	3, 1	0, 0
	d	0, 0	1, 3



		B	
		l	r
A	u	3, 1	0, 0
	d	0, 0	1, 3

		B	
		l	r
	Out	2, 1	2, 1
A	u	3, 1	0, 0
	d	0, 0	1, 3

		B	
		l	r
	Out	2, 1	2, 1
A	u	3, 1	0, 0
	d	0, 0	1, 3

		B	
		l	r
	Out	2, 1	2, 1
A	u	3, 1	0, 0
	d	0, 0	1, 3

		B	
		l	r
	Out	2, 1	2, 1
A	u	3, 1	0, 0
	d	0, 0	1, 3

		B	
		l	r
	Out	2, 1	2, 1
A	u	3, 1	0, 0
	d	0, 0	1, 3

Ann

B N

Bob

Ann	l	r
u	2, 1	-2, 0
d	-2, 0	-1, 4

Bob

Ann	l	r
u	4, 1	0, 0
d	0, 0	1, 4

Ann

B N

Bob

Ann	l	r
u	2, 1	-2, 0
d	-2, 0	-1, 4

Bob

Ann	l	r
u	4, 1	0, 0
d	0, 0	1, 4

What is forward induction reasoning?

Forward Induction Principle: a player should use all information she acquired about her opponents' past behavior in order to improve her prediction of their future simultaneous and past (unobserved) behavior, relying on the assumption that they are rational.

P. Battigalli. *On Rationalizability in Extensive Games*. Journal of Economic Theory, 74, pgs. 40 - 61, 1997.

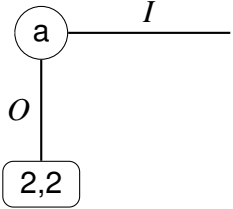
Back to our paper...

In this paper, we develop a Bayesian model of deliberation in extensive games.

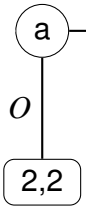
Back to our paper...

In this paper, we develop a Bayesian model of deliberation in extensive games. Our main objective is to characterize backward and forward induction reasoning in terms of the assumptions that players make about the context of the game and the update mechanisms that drive the players' deliberations.

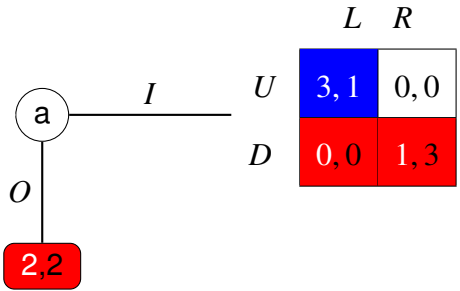
	L	R
U	$3, 1$	$0, 0$
D	$0, 0$	$1, 3$

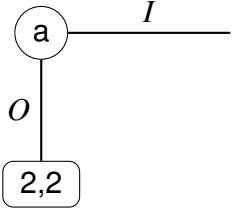


	L	R
U	$3, 1$	$0, 0$
D	$0, 0$	$1, 3$

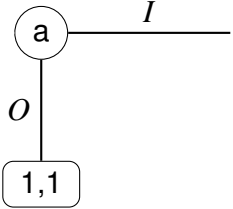


	L	R
U	$3, 1$	$0, 0$
D	$0, 0$	$1, 3$

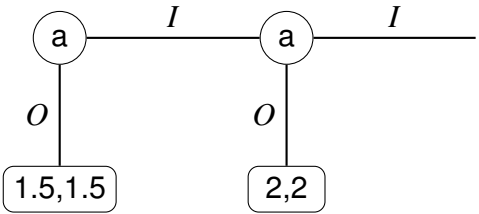

a
I
O
2,2
L
R
U
D
3, 1
0, 0
0, 0
1, 3



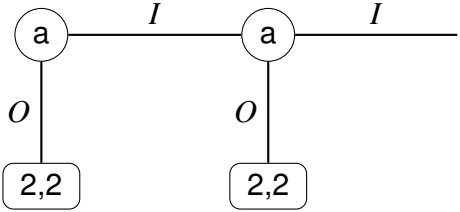
	L	R
U	$3, 1$	$0, 0$
D	$0, 0$	$1, 3$



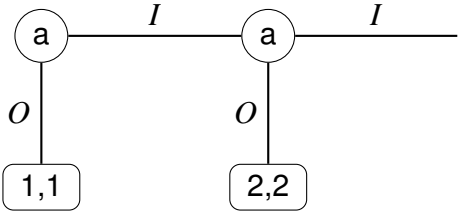
	L	R
U	$3,1$	$0,0$
D	$0,0$	$1,3$



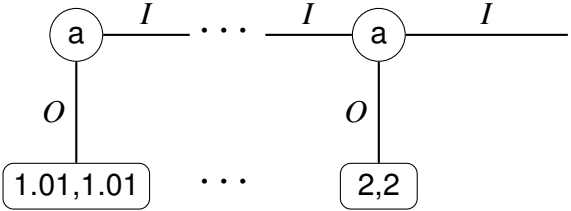
	L	R
U	$3, 1$	$0, 0$
D	$0, 0$	$1, 3$



	L	R
U	$3, 1$	$0, 0$
D	$0, 0$	$1, 3$

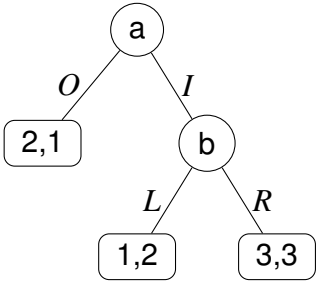


	L	R
U	$3,1$	$0,0$
D	$0,0$	$1,3$

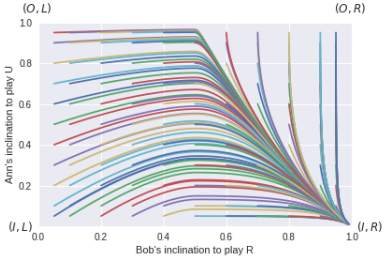


	L	R
U	$3,1$	$0,0$
D	$0,0$	$1,3$

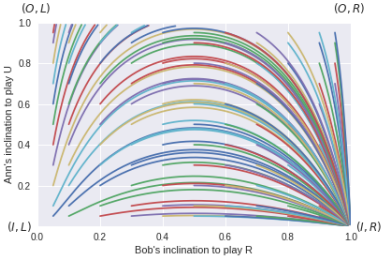
Deliberation on extensive games



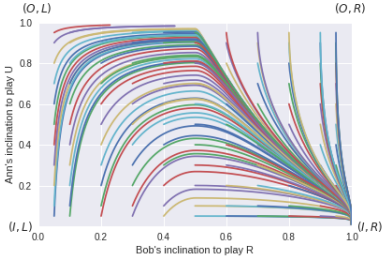
	L	R
O	$2, 1$	$2, 1$
I	$1, 2$	$3, 3$



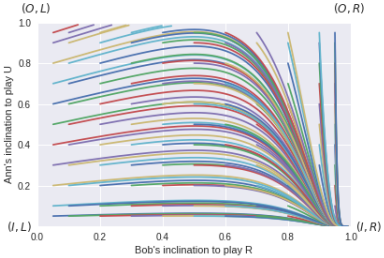
Ann is Nash, Bob is Nash



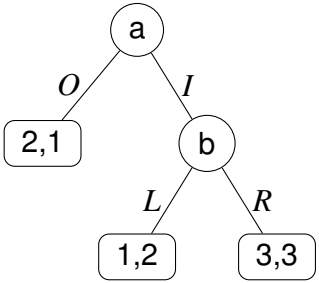
Ann is Bayes, Bob is Bayes



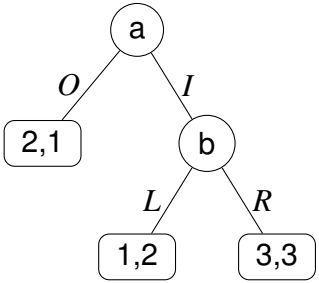
Ann is Nash, Bob is Bayes



Ann is Bayes, Bob is Nash

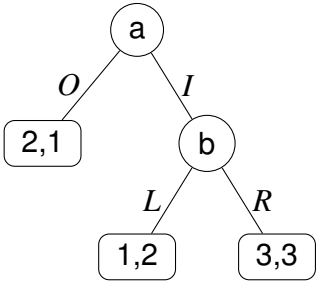


	L	R
O	$2, 1$	$2, 1$
I	$1, 2$	$3, 3$



	L	R
O	$2, 1$	$2, 1$
I	$1, 2$	$3, 3$

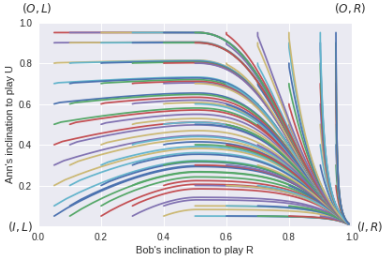
When the players deliberate simultaneously, Bob's expected utility of L is a weighted average of his payoff if Ann chooses O and his payoff if Ann chooses I.



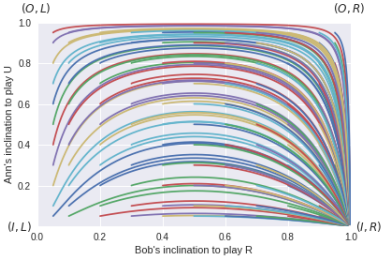
	L	R
O	$2, 1$	$2, 1$
I	$1, 2$	$3, 3$

When the players deliberate simultaneously, Bob's expected utility of L is a weighted average of his payoff if Ann chooses O and his payoff if Ann chooses I.

However, when deliberating on the extensive form game, Bob should calculate the expected utilities by conditioning on his information at his decision node: Bob should assign probability 0 to Ann choosing O, and this does not change during deliberation.



Ann is Nash, Bob is Nash



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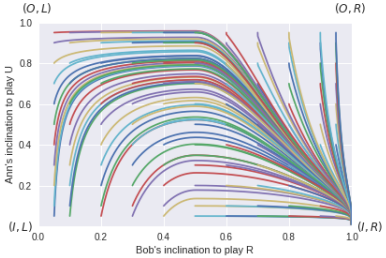
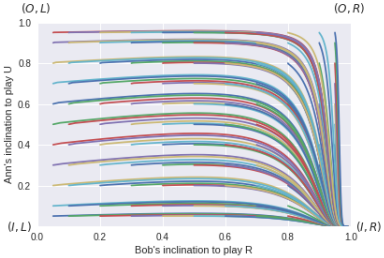


Ann is Nash, Bob is Bayes



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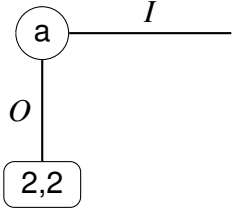
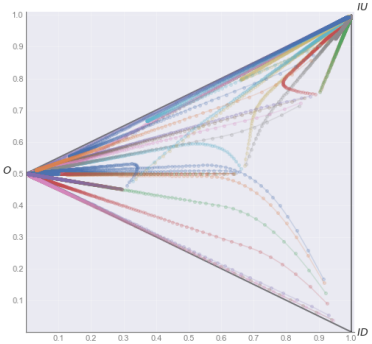
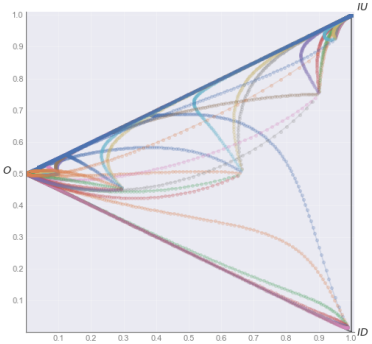


	L	R
U	$3,1$	$0,0$
D	$0,0$	$1,3$

	L	R
O	$2,2$	$2,2$
IU	$3,1$	$0,0$
ID	$0,0$	$1,3$



Nash deliberators



Bayes deliberators

Note that both Bayes and Nash deliberators converge on (IU, L) and (O, ID).

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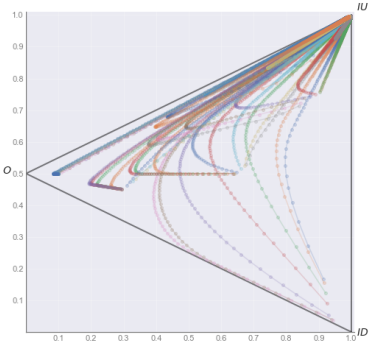
If Bob is a backward induction reasoner, then he ignores Ann's initial move as he deliberates between L and R.

Note that both Bayes and Nash deliberators converge on (IU, L) and (O, ID).

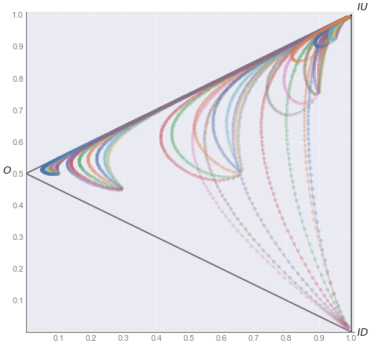
If Bob is a backward induction reasoner, then he ignores Ann's initial move as he deliberates between L and R.

On the other hand, if Bob is a forward induction reasoner, then, during deliberation, he should assign probability 0 to Ann choosing I then D (since it is strictly dominated by choosing O). This belief about Ann's choice does not change during deliberation.

Bob is a FI reasoner



Nash deliberators



Bayes deliberators

Ann may be uncertain about whether Bob is a forward induction reasoner or a backward induction reasoner (and Bob may assume this about Ann).

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Ann's belief at time $t + 1$ that Bob will play strategy s is:

$$P_a^{t+1}(s) = w_{a,t} I_{b,BI}^t(s) + (1 - w_{a,t}) I_{b,FI}^t(s)$$

where $I_{b,BI}^t$ is Bob's inclinations as a backward induction reasoner at step t, $I_{b,FI}^t$ is Bob's inclinations as a backward induction reasoner at step t, and $w_{a,t}$ is the weight Ann assigns to Bob being a backward induction reasoner at step t.

By varying the weights, we can represent players that are not fully backward induction reasoners or fully forward induction reasoners.

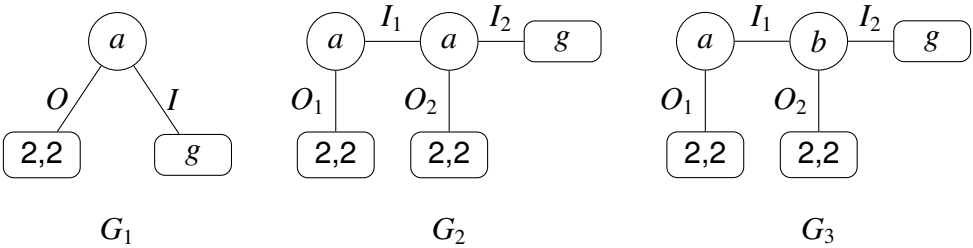
By varying the weights, we can represent players that are not fully backward induction reasoners or fully forward induction reasoners.

1. The weights $w_{a,t}$ are Ann's prior beliefs about whether Bob is a backward induction reasoner or a forward induction reasoner.

2. The weights $w_{a,t}$ are defined by the **context** of the target game.

3. The weights $w_{a,t}$ depend both on the context of the game and the probability that the players are prone to *trembling-hand mistakes*.

A context of a strategic form game G is an extensive form game G' (of perfect information) with a map g assignment G to a terminal node. If G' is a context and G is a strategic form game, then (G', G) forms an extensive game with simultaneous moves.



Concluding remarks

Our paper develops more sophisticated deliberators:

“There is nothing in the nature of deliberational dynamics that requires that deliberators be simpleminded, but the illustrations I have chosen...are relatively unsophisticated. These players follow their noses in the direction of the current apparent good, with no real memory of where they have been, no capability of recognizing patterns, and no sense of where they are going.” (Skyrms, pg. 152)

Concluding remarks

To apply the model deliberation to extensive form games, represent the informational structure of the game in the pre-deliberational beliefs, or think of deliberation in two stages: Start by allowing players to iteratively remove strictly dominated strategies, then use Nash or Bayes dynamics on the reduced game.

Concluding remarks

To apply the model deliberation to extensive form games, represent the informational structure of the game in the pre-deliberational beliefs, or think of deliberation in two stages: Start by allowing players to iteratively remove strictly dominated strategies, then use Nash or Bayes dynamics on the reduced game.

This paper takes a different perspective: The players beliefs about her opponents are a weighted average between assuming her opponent is a backward induction reasoner and a forward induction reasoner.

Concluding remarks

- Radical uncertainty about opponent's beliefs and inclinations vs. rational deliberation with more than two players

J. McKenzie Alexander. *Local Interactions and the Dynamics of Rational Deliberation*. Philosophica
Studies, 147, pgs. 102 - 121, 2010.

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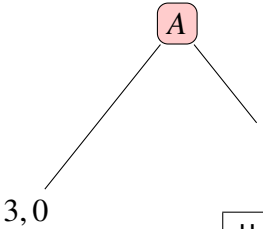
- Nature of forward induction reasoning: Does FI reasoning violate act-state dependence?

- Translucent players: reasoning in games is assumed to be opaque, but we are considering cases in which the reasoning is transparent.

  J. Halpern and R. Pass. *Game theory with translucent players*. International Journal of Gam Theory, 47:3, pgs. 949 - 976, 2018.

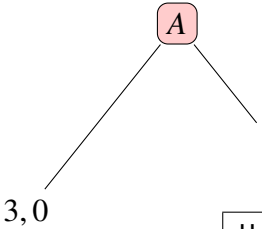
Concluding remarks: Backward *versus* Forward Induction



		Bob		
		l	c	r
Ann	u	2, 2	2, 1	0, 0
	d	1, 1	1, 2	4, 0

A. Perea. *Backward Induction* versus *Forward Induction Reasoning*. Games, 1, pgs. 168 - 188, 2010.

Concluding remarks: Backward *versus* Forward Induction



		Bob		
		l	c	r
Ann	u	2, 2	2, 1	0, 0
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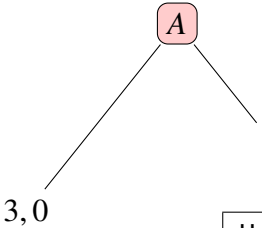

Bob

		l	c	r
Ann	u	2, 2	2, 1	0, 0
	d	1, 1	1, 2	4, 0

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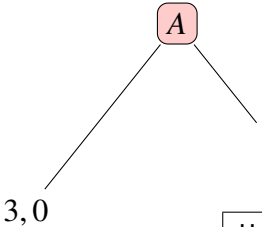

Bob

Ann	l	c	r
u	2, 2	2, 1	0, 0
d	1, 1	1, 2	4, 0

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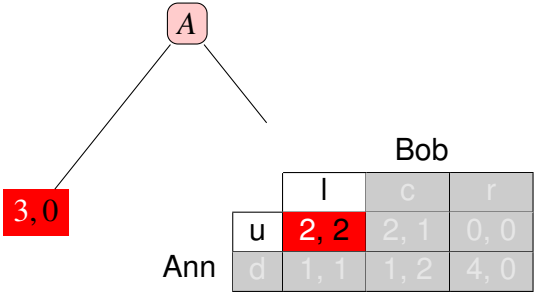

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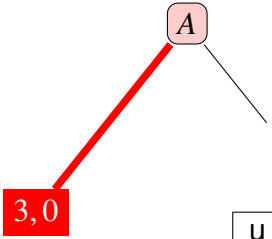

		Bob		
		l	c	r
Ann	u	2, 2	2, 1	0, 0
	d	1, 1	1, 2	4, 0

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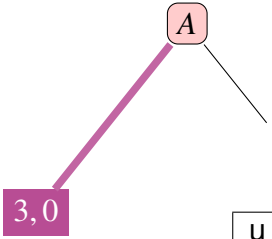


		Bob		
		l	c	r
Ann	u	2, 2	2, 1	0, 0
	d	1, 1	1, 2	4, 0

A. Perea. *Backward Induction* versus *Forward Induction Reasoning*. Games, 1, pgs. 168 - 188, 2010.

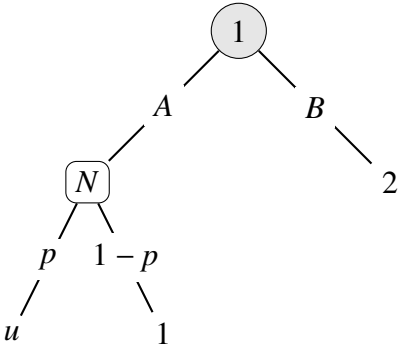
Concluding remarks: Bayesian vs. strategic rationality

M. Mariotti. *Is Bayesian Rationality Compatible with Strategic Rationality?*. The Economic Journal, 105: 432, pgs. 1099 - 1109, 1995.

M. Mariotti. *Decisions in games: why there should be a special exemption from Bayesian rationality*. Journal of Economic Methodology, 4: 1, pgs. 43 - 60, 1997.

P. Hammond. *Expected Utility in Non-Cooperative Game Theory*. in *Handbook of Utility Theory*, 2004.

Concluding remarks: Bayesian vs. strategic rationality



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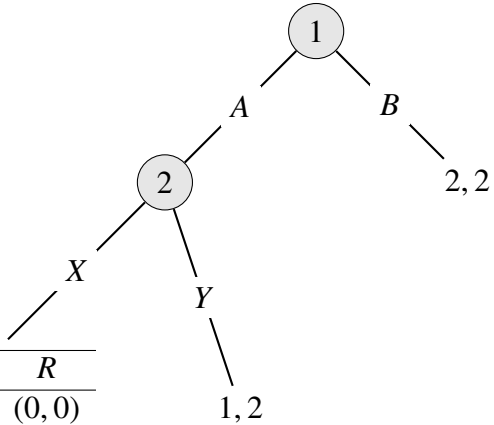

	L	R
U	$(1, 7)$	$(0, 0)$
D	$(0, 0)$	$(3, 3)$

Thank You!

epacuit@umd.edu

The Guessing Game

Guess a number between 1 & 100.
The closest to 2/3 of the average wins.

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Guess a number between 1 & 100.
The closest to 2/3 of the average wins.

What number should you guess?

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Guess a number between 1 & 100.
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What number should you guess? 100

The Guessing Game

Guess a number between 1 & 100.
The closest to 2/3 of the average wins.

What number should you guess? ~~100~~, 99

The Guessing Game

Guess a number between 1 & 100.
The closest to 2/3 of the average wins.

What number should you guess? ~~100~~, ~~99~~, . . . , 67

The Guessing Game

Guess a number between 1 & 100.
The closest to 2/3 of the average wins.

What number should you guess? ~~100~~, ~~99~~, . . . , ~~67~~,. . . , 2, 1

The Guessing Game

Guess a number between 1 & 100.
The closest to 2/3 of the average wins.

What number should you guess? $\cancel{100}$, $\cancel{99}$, . . . , $\cancel{67}$, . . . , $\cancel{2}$, 1

Traveler's Dilemma

1. You and your friend write down an integer between 2 and 100 (without discussing).

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2. If both of you write down the same number, then both will receive that amount in dollars from the airline in compensation.

Traveler's Dilemma

1. You and your friend write down an integer between 2 and 100 (without discussing).
2. If both of you write down the same number, then both will receive that amount in dollars from the airline in compensation.
3. If the numbers are different, then the airline assumes that the smaller number is the actual price of the luggage.

Traveler's Dilemma

1. You and your friend write down an integer between 2 and 100 (without discussing).
2. If both of you write down the same number, then both will receive that amount in dollars from the airline in compensation.
3. If the numbers are different, then the airline assumes that the smaller number is the actual price of the luggage.
4. The person that wrote the smaller number will receive that amount plus $2 (as a reward), and the person that wrote the larger number will receive the smaller number minus $2 (as a punishment).

Traveler's Dilemma

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4. The person that wrote the smaller number will receive that amount plus $2 (as a reward), and the person that wrote the larger number will receive the smaller number minus $2 (as a punishment).

Suppose that you are randomly paired with another person. What number would you write down?

		Bob	
		L	R
Ann	U	2,2	4,1
	D	1,4	3,3

Game 1

		Bob	
		L	R
Ann	U	2,1	1,0
	D	1,0	0,1

Game 2

		Bob	
		L	R
Ann	U	2,2	4,1
	D	1,4	3,3

Game 1

		Bob	
		L	R
Ann	U	2,1	1,0
	D	1,0	0,1

Game 2

Game 1: U strictly dominates D and L strictly dominates R.

		Bob	
		L	R
Ann	U	2,2	4,1
	D	1,4	3,3

Game 1

		Bob	
		L	R
Ann	U	2,1	1,0
	D	1,0	0,1

Game 2

Game 1: U strictly dominates D and L strictly dominates R.

Game 2: U strictly dominates D

		Bob	
		L	R
Ann	U	2,2	4,1
	D	1,4	3,3

Game 1

		Bob	
		L	R
Ann	U	2,1	1,0
	D	1,0	0,1

Game 2

Game 1: U strictly dominates D and L strictly dominates R.

Game 2: U strictly dominates D, and *after removing* D, L strictly dominates R.

		Bob	
		L	R
Ann	U	2,2	4,1
	D	1,4	3,3

Game 1

		Bob	
		L	R
Ann	U	2,1	1,0
	D	1,0	0,1

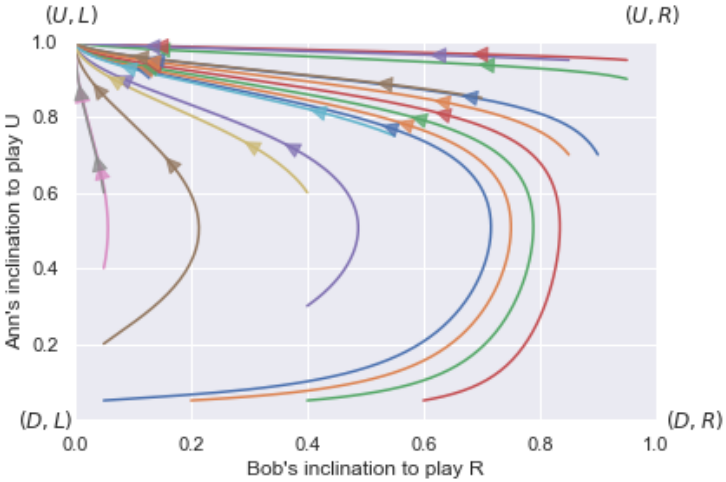
Game 2

Game 1: U strictly dominates D and L strictly dominates R.

Game 2: U strictly dominates D, and *after removing* D, L strictly dominates R.

Theorem. In all models where the players are *rational* and there is *common belief of rationality*, the players choose strategies that survive iterative removal of strictly dominated strategies (and, conversely...).

Iterated Strict Dominance - Bayes deliberators



		Bob	
		L	R
Ann	U	3,3	1,1
	D	2,2	2,2

Is R rationalizable?

		Bob	
		L	R
Ann	U	3,3	1,1
	D	2,2	2,2

Is R rationalizable?

There is no *full support* probability such that R is a best response

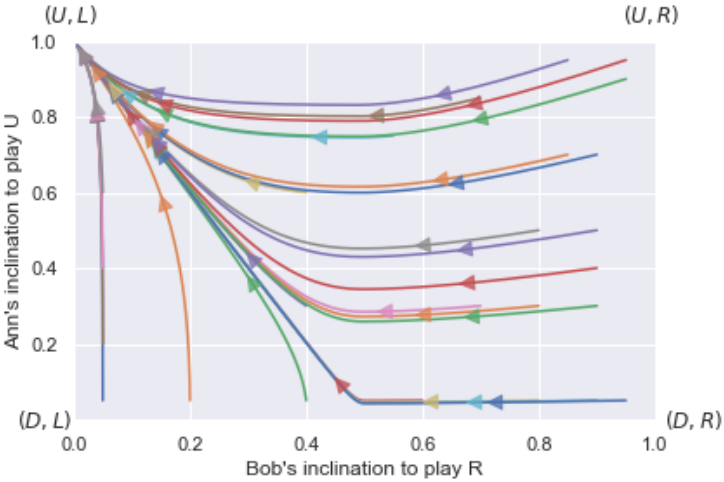
		Bob	
		L	R
Ann	U	3,3	1,1
	D	2,2	2,2

Is R rationalizable?

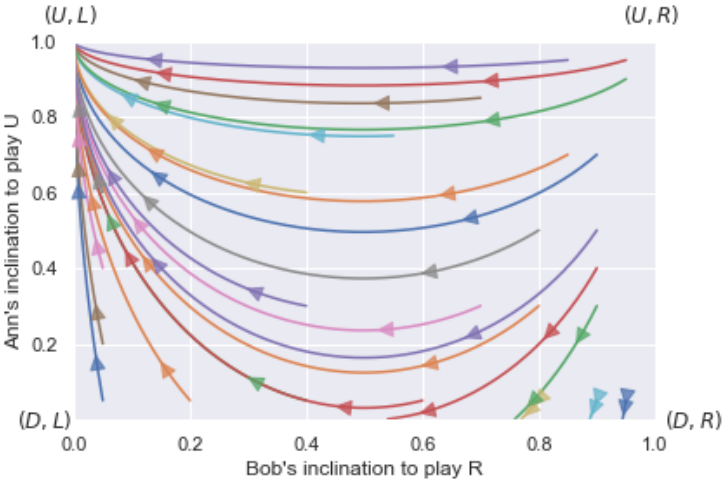
There is no *full support* probability such that R is a best response

Should Ann assign probability 0 to R or probability > 0 to R?

Weak dominance - Nash deliberators



Weak dominance - Bayes deliberators



Strategic Reasoning and Admissibility

"The argument for deletion of a weakly dominated strategy for player i is that he contemplates the possibility that every strategy combination of his rivals occurs with positive probability. However, this hypothesis clashes with the logic of iterated deletion, which assumes, precisely, that eliminated strategies are not expected to occur."

Mas-Colell, Whinston and Green. *Introduction to Microeconomics*. 1995.

A Puzzle

R. Cubitt and R. Sugden. *Rationally Justifiable Play and the Theory of Non-cooperative games.* Economic Journal, 104, pgs. 798 - 803, 1994.

R. Cubitt and R. Sugden. *Common reasoning in games: A Lewisian analysis of common knowledge of rationality*. Manuscript, 2011.

		Bob	
		L	R
Ann	U	1,1	0,0
	D	0,0	0,0

Game 1

		Bob	
		L	R
Ann	U	1,1	1,0
	D	1,0	0,1

Game 2

		Bob	
		L	R
Ann	U	1,1	0,0
	D	0,0	0,0

Game 1

		Bob	
		L	R
Ann	U	1,1	1,0
	D	1,0	0,1

Game 2

Game 1: U weakly dominates D and L weakly dominates R.

		Bob	
		L	R
Ann	U	1,1	0,0
	D	0,0	0,0

Game 1

		Bob	
		L	R
Ann	U	1,1	1,0
	D	1,0	0,1

Game 2

Game 1: U weakly dominates D and L weakly dominates R.

Game 2: U weakly dominates D

		Bob	
		L	R
Ann	U	1,1	0,0
	D	0,0	0,0

Game 1

		Bob	
		L	R
Ann	U	1,1	1,0
	D	1,0	0,1

Game 2

Game 1: U weakly dominates D and L weakly dominates R.

Game 2: U weakly dominates D, and *after removing* D, L strictly dominates R.

		Bob	
		L	R
Ann	U	1,1	0,0
	D	0,0	0,0

Game 1

		Bob	
		L	R
Ann	U	1,1	1,0
	D	1,0	0,1

Game 2

Game 1: U weakly dominates D and L weakly dominates R.

Game 2: But, now what is the reason for not playing D?

		Bob	
		L	R
Ann	U	1,1	0,0
	D	0,0	0,0

Game 1

		Bob	
		L	R
Ann	U	1,1	1,0
	D	1,0	0,1

Game 2

Game 1: U weakly dominates D and L weakly dominates R.

Game 2: But, now what is the reason for not playing D?

Theorem (Samuelson). There is no model of Game 2 satisfying common knowledge of rationality (where rationality incorporates weak dominance).

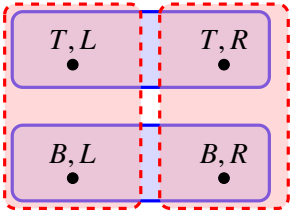
Common Knowledge of Admissibility

		Bob	
		L	R
Ann	T	1,1	1,0
	B	1,0	0,1

There is no model of this game with *common knowledge* of admissibility.

Common Knowledge of Admissibility

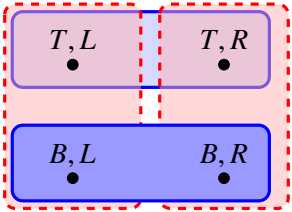
		Bob	
		L	R
Ann	T	1,1	1,0
	B	1,0	0,1



The ”full” model of the game

Common Knowledge of Admissibility

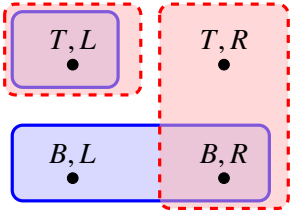
Ann \ Bob	L	R
T	1,1	1,0
B	1,0	0,1



The "full" model of the game: B *is not admissible given Ann's information*

Common Knowledge of Admissibility

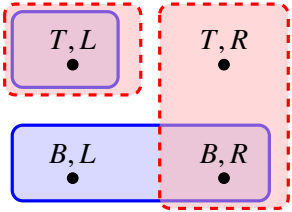
Ann \ Bob	L	R
T	1,1	1,0
B	1,0	0,1



What is wrong with this model?

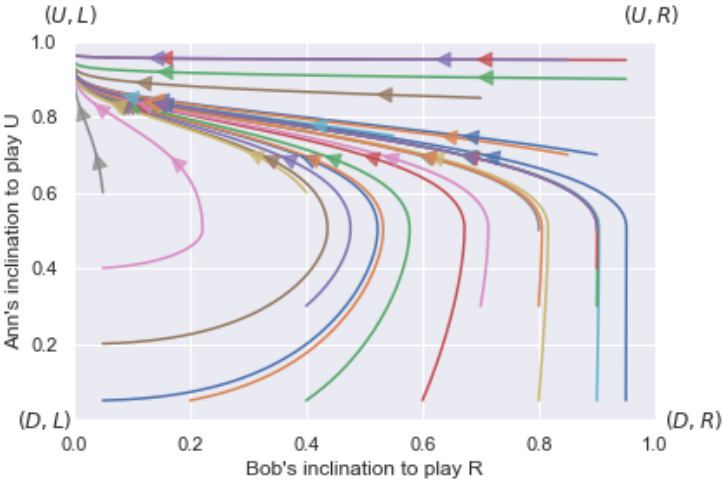
Common Knowledge of Admissibility

		Bob	
		L	R
Ann	T	1,1	1,0
	B	1,0	0,1

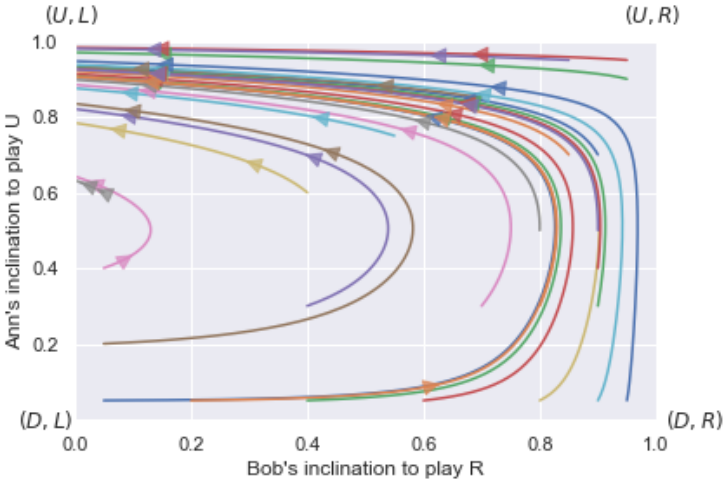


Privacy of Tie-Breaking/No Extraneous Beliefs: If a strategy is *rational* for an opponent, then it cannot be “ruled out”.

Samuelson game - Nash deliberators



Samuelson game - Bayes deliberators



Materially Rational: A player i is materially rational at a state w if every choice actually made is rational.

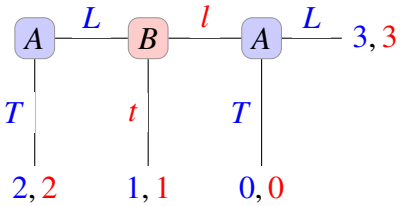
Substantively Rational: A player i is substantively rational at a state w if the player is materially rational and, in addition, for each *possible* choice, the player *would* have chosen rationally if she had had the opportunity to choose.

Materially Rational: A player i is materially rational at a state w if every choice actually made is rational.

Substantively Rational: A player i is substantively rational at a state w if the player is materially rational and, in addition, for each *possible* choice, the player *would* have chosen rationally if she had had the opportunity to choose.

E.g., Taking keys away from someone who is drunk.

Ann \ Bob	t	l
T	2,2	2,2
LT	1,1	0,0
LL	1,1	3,3



- The backward induction solution is (LL, l)

Thank You!

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